

CSS Color-Scheme

Complete Guide: Easy English Edition

Master light and dark mode support with CSS color-scheme

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1 Introduction

The CSS `color-scheme` property tells the browser and user agents which color schemes your website supports. It enables dark mode, light mode, or both!

1.1 What is Color-Scheme?

The `color-scheme` property lets you:

- **Support Dark Mode:** Automatic dark mode styling
- **Support Light Mode:** Automatic light mode styling
- **Indicate Preference:** Tell OS which schemes you support
- **Customize Elements:** Browser form elements adapt
- **Improve Accessibility:** Better for user comfort

1.2 Why Use Color-Scheme?

- **Dark Mode Support:** Free dark mode for free
- **User Preference:** Respects OS/browser preferences
- **Better UI:** Form elements look correct
- **Less CSS:** Reduces code needed
- **Accessibility:** Better for users with light sensitivity
- **Modern Standard:** Part of CSS specifications

2 Color-Scheme Basics

2.1 Basic Syntax

```
color-scheme: light;  
color-scheme: dark;  
color-scheme: light dark;
```

2.1.1 Available Values

Value	Meaning
light	Only light mode
dark	Only dark mode
light dark	Light preferred, dark supported
dark light	Dark preferred, light supported
only light	Only light (no system preference)

2.2 Simple Example

```
:root {  
  color-scheme: light dark;  
}  
  
body {  
  background-color: white;  
  color: black;  
}  
  
@media (prefers-color-scheme: dark) {  
  body {  
    background-color: #1e1e1e;  
    color: white;  
  }  
}
```

3 Implementing Color-Scheme

3.1 At Root Level

```
:root {  
  color-scheme: light dark;  
}  
  
/* All elements inherit this */
```

3.2 At HTML Tag

```
html {  
  color-scheme: light dark;  
}  
  
/* Browser knows to apply color scheme */
```

3.3 In HTML Meta Tag

```
<!-- Tells browser early, before CSS loads -->  
<meta name="color-scheme" content="light dark">
```

3.4 At Element Level

```
:root {  
  color-scheme: light dark;  
}  
  
/* Override for specific section */  
.dark-section {  
  color-scheme: dark;  
}  
  
.light-section {  
  color-scheme: light;  
}
```

4 Using prefers-color-scheme

4.1 Media Query Syntax

```
@media (prefers-color-scheme: light) {  
  /* Styles for light mode */  
}  
  
@media (prefers-color-scheme: dark) {  
  /* Styles for dark mode */  
}
```

4.2 Light Mode Styles

```
@media (prefers-color-scheme: light) {  
  body {  
    background-color: white;  
    color: black;  
  }  
  
  a {  
    color: #0066cc;  
  }  
  
  button {  
    background-color: #f0f0f0;  
    color: black;  
  }  
}
```

4.3 Dark Mode Styles

```
@media (prefers-color-scheme: dark) {  
  body {  
    background-color: #1e1e1e;  
    color: #e0e0e0;  
  }  
  
  a {  
    color: #66ccff;  
  }  
  
  button {  
    background-color: #333;  
    color: #e0e0e0;  
  }  
}
```



```
}  
}
```

5 Real-World Examples

5.1 Example 1: Simple Light/Dark Toggle

```
/* HTML Meta Tag */
<meta name="color-scheme" content="light dark">

/* CSS */
:root {
  color-scheme: light dark;
}

body {
  background-color: white;
  color: black;
  transition: background-color 0.3s, color 0.3s;
}

@media (prefers-color-scheme: dark) {
  body {
    background-color: #1e1e1e;
    color: #f0f0f0;
  }
}

/* Inputs auto-adapt */
input, textarea, select {
  background-color: inherit;
  color: inherit;
  border: 1px solid #ccc;
}

@media (prefers-color-scheme: dark) {
  input, textarea, select {
    border-color: #666;
  }
}
```

5.2 Example 2: Complete Website

```
:root {
  color-scheme: light dark;

  /* Light mode variables */
  --bg-primary: white;
  --bg-secondary: #f5f5f5;
  --text-primary: black;
```

```

    --text-secondary: #666;
    --border-color: #e0e0e0;
}

@media (prefers-color-scheme: dark) {
  :root {
    --bg-primary: #1e1e1e;
    --bg-secondary: #2d2d2d;
    --text-primary: #f0f0f0;
    --text-secondary: #b0b0b0;
    --border-color: #444;
  }
}

body {
  background-color: var(--bg-primary);
  color: var(--text-primary);
}

.card {
  background-color: var(--bg-secondary);
  color: var(--text-primary);
  border: 1px solid var(--border-color);
}

button {
  background-color: #2196F3;
  color: white;
}

input {
  background-color: var(--bg-secondary);
  color: var(--text-primary);
  border: 1px solid var(--border-color);
}

```

5.3 Example 3: Component Library

```

:root {
  color-scheme: light dark;
}

/* Button component */
.button {
  padding: 10px 20px;
  border-radius: 4px;
  border: none;
  cursor: pointer;
  transition: all 0.3s;
}

.button-primary {
  background-color: #2196F3;
  color: white;
}

```

```

.button-primary:hover {
  background-color: #1976D2;
}

@media (prefers-color-scheme: dark) {
  .button-secondary {
    background-color: #333;
    color: #e0e0e0;
  }

  .button-secondary:hover {
    background-color: #555;
  }
}

/* Card component */
.card {
  background-color: white;
  color: black;
  padding: 20px;
  border-radius: 8px;
  box-shadow: 0 2px 8px rgba(0, 0, 0, 0.1);
}

@media (prefers-color-scheme: dark) {
  .card {
    background-color: #2d2d2d;
    color: #f0f0f0;
    box-shadow: 0 2px 8px rgba(0, 0, 0, 0.3);
  }
}

```

5.4 Example 4: With Theme Toggle

```

<!DOCTYPE html>
<html>
<head>
  <meta name="color-scheme" content="light dark">
  <style>
    :root {
      color-scheme: light dark;
    }

    body {
      background: white;
      color: black;
      transition: background 0.3s, color 0.3s;
    }

    @media (prefers-color-scheme: dark) {
      body {
        background: #1e1e1e;
        color: #f0f0f0;
      }
    }
  </style>

```

```
.theme-toggle {
  position: fixed;
  top: 20px;
  right: 20px;
  background: #2196F3;
  color: white;
  border: none;
  padding: 10px 20px;
  border-radius: 4px;
  cursor: pointer;
}
</style>
</head>
<body>
  <button class="theme-toggle" onclick="toggleTheme()">
    Toggle Theme
  </button>

  <script>
    function toggleTheme() {
      const html = document.documentElement;
      const scheme = window.matchMedia('(prefers-color-scheme:
        dark)').matches
        ? 'light'
        : 'dark';
      // Manual override (would need localStorage)
    }
  </script>
</body>
</html>
```

6 Form Elements and Color-Scheme

6.1 Browser Default Behavior

```
/* With color-scheme: light dark */
input {
  /* Browser applies light or dark styles automatically */
}

button {
  /* Auto-adapts to color scheme */
}

textarea {
  /* Form elements look correct in both modes */
}
```

6.2 Customizing Form Elements

```
input {
  background-color: inherit;
  color: inherit;
  border: 1px solid currentColor;
  padding: 8px;
  border-radius: 4px;
}

@media (prefers-color-scheme: dark) {
  input {
    border-color: #666;
  }
}

button {
  background-color: #2196F3;
  color: white;
  border: none;
  padding: 10px 20px;
  border-radius: 4px;
  cursor: pointer;
}

@media (prefers-color-scheme: dark) {
  button {
    background-color: #1976D2;
  }
}
```

6.3 Color-Scheme for Inputs

```
/* Light inputs in light mode */
input[type="text"] {
  background: white;
  color: black;
  border: 1px solid #ccc;
}

/* Dark inputs in dark mode */
@media (prefers-color-scheme: dark) {
  input[type="text"] {
    background: #333;
    color: white;
    border: 1px solid #666;
  }
}

/* Scrollbar adapts too */
@media (prefers-color-scheme: dark) {
  ::-webkit-scrollbar {
    background-color: #2d2d2d;
  }

  ::-webkit-scrollbar-track {
    background-color: #1e1e1e;
  }

  ::-webkit-scrollbar-thumb {
    background-color: #666;
  }
}
```

7 Advanced Techniques

7.1 Nested Color Schemes

```
:root {
  color-scheme: light dark;
}

/* Dark section in light mode */
.dark-widget {
  color-scheme: dark;
  background: #1e1e1e;
  color: #f0f0f0;
  padding: 20px;
}

/* Light section in dark mode */
.light-widget {
  color-scheme: light;
  background: white;
  color: black;
  padding: 20px;
}
```

7.2 With Custom Properties

```
:root {
  color-scheme: light dark;

  --color-background: white;
  --color-foreground: black;
  --color-accent: #2196F3;
}

@media (prefers-color-scheme: dark) {
  :root {
    --color-background: #1e1e1e;
    --color-foreground: #f0f0f0;
    --color-accent: #66ccff;
  }
}

body {
  background: var(--color-background);
  color: var(--color-foreground);
}
```



```
a {  
  color: var(--color-accent);  
}
```

7.3 JavaScript Detection

```
<script>  
// Detect current preference  
const darkMode = window.matchMedia('(prefers-color-scheme: dark)').matches;  
  
if (darkMode) {  
  console.log('Dark mode enabled');  
} else {  
  console.log('Light mode enabled');  
}  
  
// Listen for changes  
window.matchMedia('(prefers-color-scheme:  
  dark)').addEventListener('change', (e) => {  
  if (e.matches) {  
    console.log('Switched to dark mode');  
  } else {  
    console.log('Switched to light mode');  
  }  
});  
</script>
```

8 Browser Support

8.1 Compatibility

Browser	Support
Chrome	76+
Firefox	67+
Safari	12.1+
Edge	79+
Opera	63+
IE	NOT SUPPORTED

8.2 Fallback Strategy

```
/* Always provide light mode styles first */
body {
  background: white;
  color: black;
}

/* Dark mode for supporting browsers */
@media (prefers-color-scheme: dark) {
  body {
    background: #1e1e1e;
    color: #f0f0f0;
  }
}

/* CSS color-scheme for modern browsers */
:root {
  color-scheme: light dark;
}
```

9 Best Practices

9.1 Always Use Meta Tag

```
<!-- Add early, tells browser before styles load -->
<meta name="color-scheme" content="light dark">
```

9.2 Use CSS Variables

```
/* Reduces repetition and improves maintainability */
:root {
  color-scheme: light dark;
  --bg: white;
  --fg: black;
}

@media (prefers-color-scheme: dark) {
  :root {
    --bg: #1e1e1e;
    --fg: #f0f0f0;
  }
}

body {
  background: var(--bg);
  color: var(--fg);
}
```

9.3 Test Both Modes

```
/* Test light mode */
body {
  background: white;
  color: black;
}

/* Test dark mode */
@media (prefers-color-scheme: dark) {
  body {
    background: #1e1e1e;
    color: #f0f0f0;
  }
}
```

9.4 Sufficient Contrast

```
/* Light mode: dark text on light background */
body {
  background: white;
  color: #333; /* Dark enough */
}

/* Dark mode: light text on dark background */
@media (prefers-color-scheme: dark) {
  body {
    background: #1e1e1e;
    color: #e0e0e0; /* Light enough */
  }
}
```

10 Common Mistakes

10.1 Mistake 1: Forgetting Meta Tag

DON'T:

```
<!-- No meta tag -->
<!-- Only CSS declaration -->
:root {
  color-scheme: light dark;
}
```

DO:

```
<!-- Add meta tag -->
<meta name="color-scheme" content="light dark">

<!-- And CSS -->
:root {
  color-scheme: light dark;
}
```

10.2 Mistake 2: Only Supporting Dark Mode

DON'T:

```
:root {
  color-scheme: dark; /* Users can't use light mode */
}
```

DO:

```
:root {
  color-scheme: light dark; /* Both modes supported */
}
```

10.3 Mistake 3: Poor Contrast

DON'T:

```
@media (prefers-color-scheme: dark) {
  body {
    background: #2d2d2d;
    color: #333; /* Too dark! */
  }
}
```

DO:

```
@media (prefers-color-scheme: dark) {  
  body {  
    background: #1e1e1e;  
    color: #e0e0e0; /* Good contrast */  
  }  
}
```

10.4 Mistake 4: Ignoring Form Elements

DON'T:

```
/* Only styled body */  
body {  
  color-scheme: light dark;  
}  
/* Forms look broken */
```

DO:

```
/* Style everything */  
:root {  
  color-scheme: light dark;  
}  
  
input, textarea, button {  
  background: inherit;  
  color: inherit;  
}
```

11 Quick Reference

11.1 Essential CSS

```
/* 1. Meta tag in HTML */
<meta name="color-scheme" content="light dark">

/* 2. CSS at root */
:root {
  color-scheme: light dark;
}

/* 3. Light mode styles (default) */
body {
  background: white;
  color: black;
}

/* 4. Dark mode styles */
@media (prefers-color-scheme: dark) {
  body {
    background: #1e1e1e;
    color: #f0f0f0;
  }
}
```

11.2 Color-Scheme Values

color-scheme: light;	/* Light only */
color-scheme: dark;	/* Dark only */
color-scheme: light dark;	/* Both, light preferred */
color-scheme: dark light;	/* Both, dark preferred */
color-scheme: only light;	/* Light, ignore system */

11.3 Key Points

- `color-scheme` tells browser which modes you support
- Use `prefers-color-scheme` media query for styles
- Always include meta tag in HTML head
- Support both light and dark modes
- Use CSS variables to reduce repetition
- Ensure sufficient color contrast

- Test in both light and dark modes
- Browser applies defaults to form elements

Conclusion

CSS color-scheme is the foundation for proper dark mode support. Combined with prefers-color-scheme media queries and CSS variables, it enables a modern, accessible web experience that respects user preferences.

Key Takeaways:

- Add `color-scheme` meta tag early
- Use `color-scheme: light dark;` in CSS
- Apply styles with `@media (prefers-color-scheme: dark)`
- Use CSS variables to manage color themes
- Test thoroughly in both modes
- Ensure proper contrast ratios
- Support user OS/browser preferences

Start implementing color-scheme for better dark mode support today!